

Roll No.

41477

**B. Pharma 4th Semester
(Regular/Re-appear/Impr.)
Examination – May, 2024**

MEDICINAL CHEMISTRY-I

Paper : BT402T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *all* questions from Section-A & any *two* questions from Section-B and any *seven* questions from Section-C.

SECTION – A

1. Attempt *all* questions : $2 \times 10 = 20$
- (a) Define solubility.
 - (b) What is ring equivalent bioisosterism ?
 - (c) Write the mode of action of methyldopa.
 - (d) Write the synthetic route of Phenylephrine.
 - (e) Define parasympatholytic agents.

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- (f) Classify cholinesterase inhibitors.
- (g) What are the ideal characteristics of sedatives and hypnotics?
- (h) Draw the chemical structure of Clonazepam and write MOA.
- (i) Write the synthetic route of halothane.
- (j) Write the mode of action of Ultra short acting barbiturates

SECTION – B

Long Answer Questions (Attempt any *two*) : $10 \times 2 = 20$

2. Explain in detail about isosterism and bioisosterism with suitable examples.
3. Discuss the SAR beta blocker and write the mode of action synthesis of propranolol.
4. Write the chemical structure, mode of action, synthesis and use of carbachol and procyclidine.

SECTION – C

Short Answer Questions (Attempt any *Seven*) : $5 \times 7 = 35$

5. Discuss the SAR of barbiturate with suitable examples.
6. Classify anti-inflammatory agents.

Do

7. Discuss the chemical structure mode of action and synthesis of ibuprofen.

8. Describe geometrical isomerism in relation to affect biological activity.

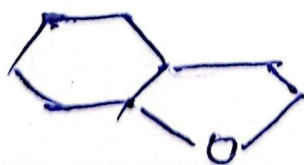
9. Discuss in detail about indirect acting sympathomimetic agents.

10. Classify anticonvulsant drugs and explain SAR of succinimide.

11. Write the mechanism of action and synthesis of ethosuccimide.

12. Discuss the SAR of morphine analogues.

13. Write the mechanism of action and synthesis of fentanyl.



Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences

B.Pharma 4th Semester, 1st Sessional Examination, 2024

Paper Code- BP402T Pharmaceutical Medicinal Chemistry (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

Question no.1 is compulsory

- a. Structure MOA and Uses of prazosin
- b. Structure MOA and Uses of phenylephrine
- c. Structure MOA and Uses of Oxymetazoline.
- d. Structure MOA and Uses of isoproterenol.
- e. Structure MOA and Uses of Labetolol

2x5=10 marks

Q2. Attempt any one questions from the rest

10x1=10 marks

- a. Biosynthesis and catabolism of catecholamine?
- b. Synthesis and uses of Tolazoline and Propranolol?

Q3. Attempt any two questions from the rest

5x2=10 marks

- a. Discuss SAR of sympathomimetic agents?
- b. writes a note on the history of medicinal chemistry? 1928
- c. Factors affecting drug metabolism including stereo chemical aspects?

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B.Pharm 4th Semester, 2nd Sessional Examination, 2024
Paper Code- BP402T Medicinal Chemistry-III (Theory)

Duration: 1 hr 30 minutes

Total marks: 30

1-Attempt all questions each question carry 2 marks.

- Differentiate between Narcotic & non narcotics analgesics .
- Mention the mechanism of action of Antipsychotics agents .
- Compare the basic ring structures and mention uses of barbiturate & benzodiazepines .
- Differentiate anticholinergics and anticholinesterases.
- Give name & structures of two narcotic antagonists .

2x5=10 marks

2 Attempt any one question .

- Classify sedative and hypnotics .
- Give the classification & SAR of Cholinergic Blocking Hydrochloride.

Outline the synthesis & MOA of Diazepam & Barbiturates

10x1=10 marks

Write down the synthesis of Dicycloamine

3 Attempt any two questions .

- Give the classification of Narcotic & non narcotics analgesics with structures .
- Discuss the classification & SAR of Parasympathomimetic agents .
- Explain the classification & SAR of Cholinergic Blocking Dicycloamine Hydrochloride.

5x2=10 marks

Write down the

SAR, Benzodiazepines, Barbiturates